Course 6032 | Semester VI | Theory | 4 Credits

Electric Vehicle and Renewable Energy Technologies

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6032 as Electric Vehicle and Renewable Energy Technologies in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Electric Vehicles and Renewable Energy course and polytechnic renewable energy curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. EV designs, battery systems, solar installations and grid connections must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support Solar PV, Wind energy, EV batteries, motors, charging infrastructure, and grid integration. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electric Vehicle and Renewable Energy Technologies studies the principles, components and integration of sustainable energy systems and electric transportation. It develops the ability to analyze renewable energy sources (Solar, Wind), evaluate energy storage systems (Batteries, Fuel Cells), understand electric vehicle powertrains and charging infrastructure, and coordinate green energy solutions while respecting the technical and safety boundaries of electrical and automotive engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electrical engineering, power electronics, electrical machines and energy conversion principles.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the principles and technologies of renewable energy sources including Solar PV and Wind energy.
2. Explain the operation, characteristics and management of energy storage systems like Li-ion batteries and Fuel Cells.
3. Analyze the architecture, components and performance of electric vehicle (EV) powertrains and motor drives.
4. Explain the infrastructure, standards and grid integration requirements for electric vehicle charging systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the working principles and components of Solar PV and Wind energy systems.	Understand
CO2	Explain the characteristics and management of batteries and energy storage for EVs.	Understand
CO3	Explain the architecture and motor drive systems used in electric vehicles.	Understand
CO4	Explain EV charging infrastructure, standards and renewable energy grid integration.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Renewable Energy Sources	Global energy scenario; Solar Photovoltaic (PV) systems: cell, module, array, MPPT; Wind Energy: turbine types, power calculation, grid connection; Biomass and Small Hydro basics.	Approx. 14 hours	CO1	Module 1
2	Energy Storage and Management	Energy storage needs; Battery technologies: Lead-acid, Li-ion, Ni-MH; Battery characteristics: SoC, SoH, C-rating; Battery Management Systems (BMS); Fuel Cells and Ultracapacitors.	Approx. 14 hours	CO2	Module 2
3	Electric Vehicle Powertrain and Drives	EV configurations: BEV, HEV, PHEV, FCEV; Powertrain components; EV Motors: BLDC, PMSM, Induction Motor; Motor controllers and Power Electronics converters.	Approx. 16 hours	CO3	Module 3
4	Charging Infrastructure and Grid Integration	Charging levels (Level 1, 2, 3); Charging standards: CHAdeMO, CCS, GB/T; On-board and Off-board chargers; Vehicle-to-Grid (V2G) concepts; Renewable energy integration and Smart Grids.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Renewable Energy Sources

Global energy scenario; Solar Photovoltaic (PV) systems: cell, module, array, MPPT; Wind Energy: turbine types, power calculation, grid connection; Biomass and Small Hydro basics.

CO1 · Approx. 14 hours

Energy Storage and Management

Energy storage needs; Battery technologies: Lead-acid, Li-ion, Ni-MH; Battery characteristics: SoC, SoH, C-rating; Battery Management Systems (BMS); Fuel Cells and Ultracapacitors.

CO2 · Approx. 14 hours

Electric Vehicle Powertrain and Drives

EV configurations: BEV, HEV, PHEV, FCEV; Powertrain components; EV Motors: BLDC, PMSM, Induction Motor; Motor controllers and Power Electronics converters.

CO3 · Approx. 16 hours

Charging Infrastructure and Grid Integration

Charging levels (Level 1, 2, 3); Charging standards: CHAdeMO, CCS, GB/T; On-board and Off-board chargers; Vehicle-to-Grid (V2G) concepts; Renewable energy integration and Smart Grids.

CO4 · Approx. 16 hours

MODULE 1

Renewable Energy Sources

Global energy scenario; Solar Photovoltaic (PV) systems: cell, module, array, MPPT; Wind Energy: turbine types, power calculation, grid connection; Biomass and Small Hydro basics.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

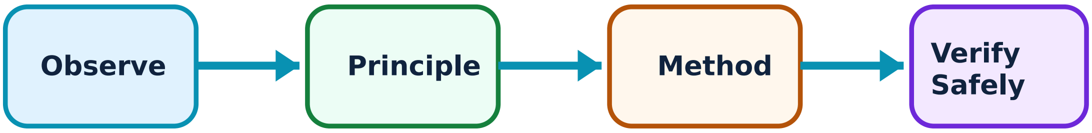
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Renewable Energy Sources: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solar Photovoltaic Systems−

Solar PV converts sunlight directly into electricity using semiconductor cells. A system includes PV modules, inverters, and Maximum Power Point Tracking (MPPT) controllers to optimize energy harvest. Grid-tied and off-grid configurations are used for various applications.

Study and application method

Describe the working of a 'Solar Cell'; explain the role of 'MPPT' in a PV system; differentiate between 'Grid-tied' and 'Stand-alone' solar systems.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of a 'Solar Cell'; explain the role of 'MPPT' in a PV system; differentiate between 'Grid-tied' and 'Stand-alone' solar systems.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Solar installations involve high-voltage DC. Do not install or maintain PV systems without authorised electrical safety equipment and adherence to local building codes.

2. Wind Energy Conversion−

Wind turbines convert kinetic energy from wind into mechanical energy, then into electricity. Horizontal Axis Wind Turbines (HAWT) are common for large-scale generation. Power output depends on wind speed, blade area, and air density.

Study and application method

Sketch the components of a 'Wind Energy Conversion System' (WECS); list the factors affecting the power output of a wind turbine.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the components of a 'Wind Energy Conversion System' (WECS); list the factors affecting the power output of a wind turbine.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wind turbines involve rotating machinery and high-altitude work. Do not approach or climb turbines without authorised safety training and fall-protection gear.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Energy Storage and Management

Energy storage needs; Battery technologies: Lead-acid, Li-ion, Ni-MH; Battery characteristics: SoC, SoH, C-rating; Battery Management Systems (BMS); Fuel Cells and Ultracapacitors.

Approx. 14 hours | **CO2** | **Understand · Apply**

Learning goals

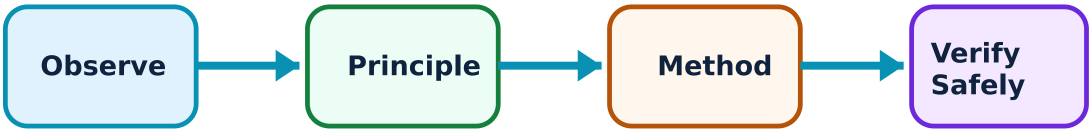
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Storage and Management: identify the system, state its principle, follow the correct method and verify the result safely.

1. Battery Technologies for EVs−

Batteries are the primary energy source for EVs. Lithium-ion (Li-ion) batteries are preferred for their high energy density and long cycle life. State of Charge (SoC) and State of Health (SoH) are critical parameters for monitoring battery status.

Study and application method

Compare 'Li-ion' and 'Lead-acid' batteries for EV applications; define 'C-rating' and its significance in battery charging/discharging.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Li-ion' and 'Lead-acid' batteries for EV applications; define 'C-rating' and its significance in battery charging/discharging.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Li-ion batteries pose fire and explosion risks if damaged or overcharged. Do not handle or charge EV batteries without authorised BMS and thermal management systems.

2. Battery Management Systems (BMS)−

BMS ensures the safe and efficient operation of battery packs. It monitors cell voltages, currents, and temperatures, performs cell balancing, and provides protection against overvoltage, undervoltage, and thermal runaway.

Study and application method

List the main functions of a 'Battery Management System'; explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the main functions of a 'Battery Management System'; explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: BMS failure can lead to catastrophic battery failure. All battery packs must use authorised BMS hardware and software verified for the specific cell chemistry.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Electric Vehicle Powertrain and Drives

EV configurations: BEV, HEV, PHEV, FCEV; Powertrain components; EV Motors: BLDC, PMSM, Induction Motor; Motor controllers and Power Electronics converters.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

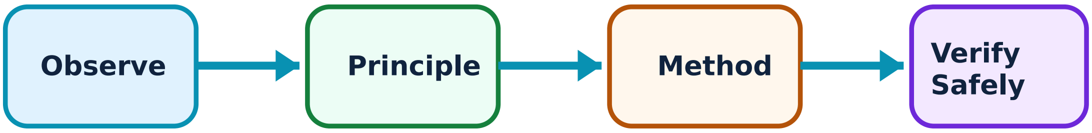
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electric Vehicle Powertrain and Drives: identify the system, state its principle, follow the correct method and verify the result safely.

1. EV Architectures and Components−

Electric vehicles come in various configurations. Battery Electric Vehicles (BEVs) rely solely on batteries. Hybrid Electric Vehicles (HEVs) combine an internal combustion engine with an electric motor. The powertrain includes the battery, controller, motor, and transmission.

Study and application method

Differentiate between 'BEV', 'HEV', and 'PHEV' with conceptual diagrams; list the key components of an EV powertrain.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'BEV', 'HEV', and 'PHEV' with conceptual diagrams; list the key components of an EV powertrain.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: EV powertrains involve high-voltage DC and AC circuits. Do not perform maintenance on EV drives without authorised high-voltage safety training and PPE.

2. Electric Motors and Control−

EVs use high-efficiency motors. BLDC (Brushless DC) and PMSM (Permanent Magnet Synchronous Motor) are popular for their high power density. Induction motors are robust and cost-effective. Motor controllers use PWM to regulate speed and torque.

Study and application method

Explain the advantages of 'PMSM' over 'Induction Motors' in EVs; describe the role of a 'Motor Controller' in an electric vehicle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the advantages of 'PMSM' over 'Induction Motors' in EVs; describe the role of a 'Motor Controller' in an electric vehicle.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect motor control parameters can cause sudden acceleration or braking. All motor drive software must be verified and follow authorised automotive safety standards.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Charging Infrastructure and Grid Integration

Charging levels (Level 1, 2, 3); Charging standards: CHAdeMO, CCS, GB/T; On-board and Off-board chargers; Vehicle-to-Grid (V2G) concepts; Renewable energy integration and Smart Grids.

Approx. 16 hours **CO4** **Understand · Apply**

Learning goals

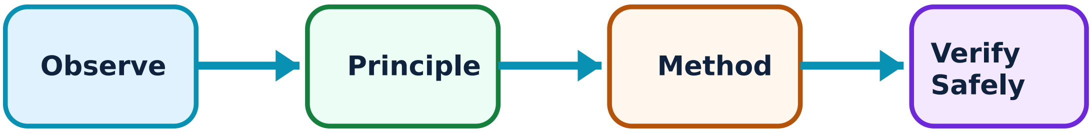
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Charging Infrastructure and Grid Integration: identify the system, state its principle, follow the correct method and verify the result safely.

1. EV Charging Standards and Systems−

Charging systems are categorized by power level and speed. Level 1 is slow (home), Level 2 is medium (public/work), and Level 3 is fast (DC fast charging). Standards like CCS and CHAdeMO ensure compatibility between vehicles and chargers.

Study and application method

Compare 'AC Charging' and 'DC Fast Charging' conceptually; list three international EV charging standards.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'AC Charging' and 'DC Fast Charging' conceptually; list three international EV charging standards.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-power charging involves significant heat and electrical stress. Do not use charging equipment that is not authorised, certified, or properly grounded.

2. Grid Integration and V2G−

V2G technology allows EVs to return energy to the power grid during peak demand, acting as mobile storage. This helps balance the grid and integrate intermittent renewable sources like solar and wind into a 'Smart Grid' environment.

Study and application method

Explain the concept of 'Vehicle-to-Grid' (V2G); describe how EVs can support the integration of renewable energy into the grid.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of 'Vehicle-to-Grid' (V2G); describe how EVs can support the integration of renewable energy into the grid.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Grid-tied EV systems must comply with utility regulations. Do not connect V2G systems without authorised grid-interface equipment and utility approval.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Renewable Energy Sources

Use the concept flow to revise observation, principle, method and verification.

Module 2: Energy Storage and Management

Use the concept flow to revise observation, principle, method and verification.

Module 3: Electric Vehicle Powertrain and Drives

Use the concept flow to revise observation, principle, method and verification.

Module 4: Charging Infrastructure and Grid Integration

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare BEV and HEV architectures in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the block diagram of a Grid-tied Solar PV system showing the PV array, MPPT, and Inverter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the energy capacity conceptually for a battery pack with a given voltage and Ah rating.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a simple flowchart showing the operation of a Battery Management System (BMS).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the EV charging stations available in your city or region using a map or app.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Renewable Energy Sources

Explain Solar Photovoltaic Systems and state one correct method, application or precaution.−

Solar PV converts sunlight directly into electricity using semiconductor cells. A system includes PV modules, inverters, and Maximum Power Point Tracking (MPPT) controllers to optimize energy harvest. Grid-tied and off-grid configurations are used for various applications.

Answer framework: Describe the working of a 'Solar Cell'; explain the role of 'MPPT' in a PV system; differentiate between 'Grid-tied' and 'Stand-alone' solar systems.

Important point: Solar installations involve high-voltage DC. Do not install or maintain PV systems without authorised electrical safety equipment and adherence to local building codes.

Explain Wind Energy Conversion and state one correct method, application or precaution.−

Wind turbines convert kinetic energy from wind into mechanical energy, then into electricity. Horizontal Axis Wind Turbines (HAWT) are common for large-scale generation. Power output depends on wind speed, blade area, and air density.

Answer framework: Sketch the components of a 'Wind Energy Conversion System' (WECS); list the factors affecting the power output of a wind turbine.

Important point: Wind turbines involve rotating machinery and high-altitude work. Do not approach or climb turbines without authorised safety training and fall-protection gear.

Module 2 — Energy Storage and Management

Explain Battery Technologies for EVs and state one correct method, application or precaution.−

Batteries are the primary energy source for EVs. Lithium-ion (Li-ion) batteries are preferred for their high energy density and long cycle life. State of Charge (SoC) and State of Health (SoH) are critical parameters for monitoring battery status.

Answer framework: Compare 'Li-ion' and 'Lead-acid' batteries for EV applications; define 'C-rating' and its significance in battery charging/discharging.

Important point: Li-ion batteries pose fire and explosion risks if damaged or overcharged. Do not handle or charge EV batteries without authorised BMS and thermal management systems.

Explain Battery Management Systems (BMS) and state one correct method, application or precaution.−

BMS ensures the safe and efficient operation of battery packs. It monitors cell voltages, currents, and temperatures, performs cell balancing, and provides protection against overvoltage, undervoltage, and thermal runaway.

Answer framework: List the main functions of a 'Battery Management System'; explain why 'Cell Balancing' is necessary in a multi-cell battery pack.

Important point: BMS failure can lead to catastrophic battery failure. All battery packs must use authorised BMS hardware and software verified for the specific cell chemistry.

Module 3 — Electric Vehicle Powertrain and Drives

Explain EV Architectures and Components and state one correct method, application or precaution.−

Electric vehicles come in various configurations. Battery Electric Vehicles (BEVs) rely solely on batteries. Hybrid Electric Vehicles (HEVs) combine an internal combustion engine with an electric motor. The powertrain includes the battery, controller, motor, and transmission.

Answer framework: Differentiate between 'BEV', 'HEV', and 'PHEV' with conceptual diagrams; list the key components of an EV powertrain.

Important point: EV powertrains involve high-voltage DC and AC circuits. Do not perform maintenance on EV drives without authorised high-voltage safety training and PPE.

Explain Electric Motors and Control and state one correct method, application or precaution.−

EVs use high-efficiency motors. BLDC (Brushless DC) and PMSM (Permanent Magnet Synchronous Motor) are popular for their high power density. Induction motors are robust and cost-effective. Motor controllers use PWM to regulate speed and torque.

Answer framework: Explain the advantages of 'PMSM' over 'Induction Motors' in EVs; describe the role of a 'Motor Controller' in an electric vehicle.

Important point: Incorrect motor control parameters can cause sudden acceleration or braking. All motor drive software must be verified and follow authorised automotive safety standards.

Module 4 — Charging Infrastructure and Grid Integration

Explain EV Charging Standards and Systems and state one correct method, application or precaution.−

Charging systems are categorized by power level and speed. Level 1 is slow (home), Level 2 is medium (public/work), and Level 3 is fast (DC fast charging). Standards like CCS and CHAdeMO ensure compatibility between vehicles and chargers.

Answer framework: Compare 'AC Charging' and 'DC Fast Charging' conceptually; list three international EV charging standards.

Important point: High-power charging involves significant heat and electrical stress. Do not use charging equipment that is not authorised, certified, or properly grounded.

Explain Grid Integration and V2G and state one correct method, application or

precaution.−

V2G technology allows EVs to return energy to the power grid during peak demand, acting as mobile storage. This helps balance the grid and integrate intermittent renewable sources like solar and wind into a 'Smart Grid' environment.

Answer framework: Explain the concept of 'Vehicle-to-Grid' (V2G); describe how EVs can support the integration of renewable energy into the grid.

Important point: Grid-tied EV systems must comply with utility regulations. Do not connect V2G systems without authorised grid-interface equipment and utility approval.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Renewable Energy Sources.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Energy Storage and Management.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Electric Vehicle Powertrain and Drives.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Charging Infrastructure and Grid Integration.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Global energy scenario; Solar Photovoltaic (PV) systems: cell, module, array, MPPT; Wind Energy: turbine types, power calculation, grid connection; Biomass and Small Hydro basics..

2. Develop a complete answer or practical plan for: Energy storage needs; Battery technologies: Lead-acid, Li-ion, Ni-MH; Battery characteristics: SoC, SoH, C-rating; Battery Management Systems (BMS); Fuel Cells and Ultracapacitors..
3. Develop a complete answer or practical plan for: EV configurations: BEV, HEV, PHEV, FCEV; Powertrain components; EV Motors: BLDC, PMSM, Induction Motor; Motor controllers and Power Electronics converters..
4. Develop a complete answer or practical plan for: Charging levels (Level 1, 2, 3); Charging standards: CHAdeMO, CCS, GB/T; On-board and Off-board chargers; Vehicle-to-Grid (V2G) concepts; Renewable energy integration and Smart Grids..

LAST-MILE REVISION

Quick Revision

Module 1: Renewable Energy Sources

Global energy scenario; Solar Photovoltaic (PV) systems: cell, module, array, MPPT; Wind Energy: turbine types, power calculation, grid connection; Biomass and Small Hydro basics.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Energy Storage and Management

Energy storage needs; Battery technologies: Lead-acid, Li-ion, Ni-MH; Battery characteristics: SoC, SoH, C-rating; Battery Management Systems (BMS); Fuel Cells and Ultracapacitors.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Electric Vehicle Powertrain and Drives

EV configurations: BEV, HEV, PHEV, FCEV; Powertrain components; EV Motors: BLDC, PMSM, Induction Motor; Motor controllers and Power Electronics converters.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Charging Infrastructure and Grid Integration

Charging levels (Level 1, 2, 3); Charging standards: CHAdeMO, CCS, GB/T; On-board and Off-board chargers; Vehicle-to-Grid (V2G) concepts; Renewable energy integration and Smart Grids.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Solar Photovoltaic Systems	Solar PV converts sunlight directly into electricity using semiconductor cells.
Wind Energy Conversion	Wind turbines convert kinetic energy from wind into mechanical energy, then into electricity.
Battery Technologies for EVs	Batteries are the primary energy source for EVs.
Battery Management Systems (BMS)	BMS ensures the safe and efficient operation of battery packs.
EV Architectures and Components	Electric vehicles come in various configurations.
Electric Motors and Control	EVs use high-efficiency motors.
EV Charging Standards and Systems	Charging systems are categorized by power level and speed.
Grid Integration and V2G	V2G technology allows EVs to return energy to the power grid during peak demand, acting as mobile storage.

LEARNING RESOURCES

References

Recommended EV and renewable energy references

1. Ashok Jhunjhunwala, Electric Vehicles and Renewable Energy.
2. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals.
3. James Larminie and John Lowry, Electric Vehicle Technology Explained.
4. Gilbert M. Masters, Renewable and Efficient Electric Power Systems.

Approved public EV and renewable energy sources

- https://nptel.ac.in/courses/108106170
- https://nptel.ac.in/courses/108107143

These sources provide the verified study basis while official Course 6032 detail is unavailable; they do not replace official vehicle designs, authorised battery safety protocols, certified solar installations or authorised grid management plans.